Project BMDC

MEDICINE · TOPIC NOTES

Hepatic & *Biliary* System

Anatomy, liver failure, portal hypertension, jaundice, hepatitis and biliary disease, organised for the BMDC licensing exam.

Project BMDC
Your gateway to BMDC licence

22 topics
BMDC RQE preparation

Contents

Read top to bottom: structure first, then disease, then complications.

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Green boxes
Key exam points: definitions, cut-offs and "most common".

Dark cards
Treatment and management.

Diagrams
Drawn to show mechanisms. Learn the arrows, not just the words.

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PART 1

Anatomy & Physiology

Liver

Largest gland of the body. **Weight: 1.2–1.5 kg** in an adult.

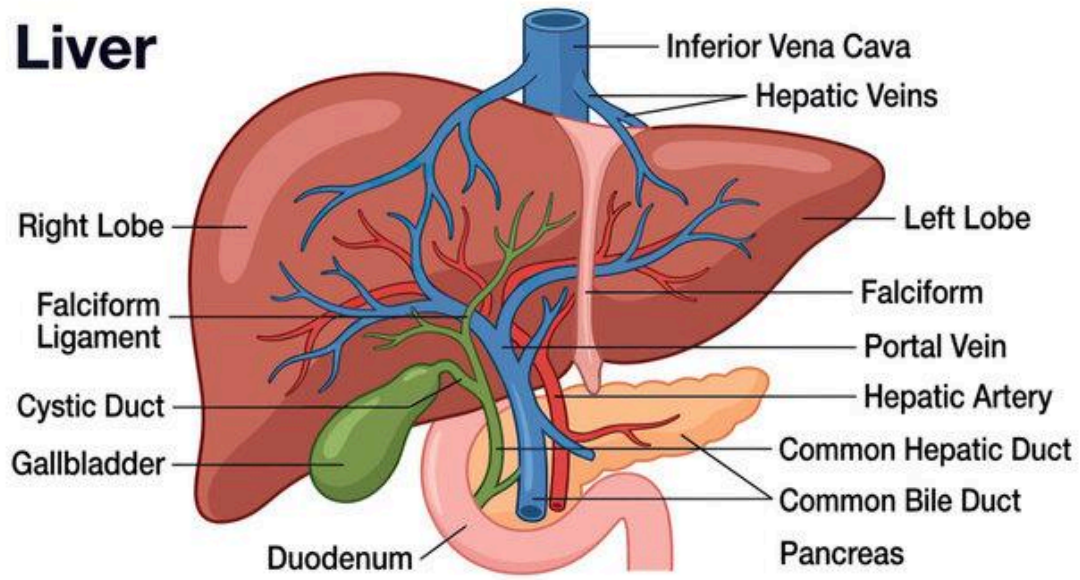


Fig 1. Liver: lobes, vessels and biliary tree.

BLOOD FLOW IN

- **Hepatic artery**
- **Portal vein**

BLOOD FLOW OUT

- **Hepatic veins** → inferior vena cava

MAJOR FUNCTIONS OF THE LIVER

1 Synthesis

- Albumin, fibrinogen, prothrombin
- Clotting factors II, VII, IX, X
- Acute phase proteins
- VLDL, cholesterol

2 Metabolism

- **Carbohydrate:** glycogenesis, glycogenolysis, gluconeogenesis
- **Protein:** transamination, deamination, urea synthesis
- **Lipid:** β-oxidation of fatty acids, ketone body formation

3 Detoxification

- Phase I (cytochrome P450)
- Phase II conjugation
- For drugs, toxins, alcohol

4 Storage

- Vitamin A, Vitamin D, B12
- Iron
- Glycogen

5 Reticuloendothelial

- Kupffer cells phagocytose bacteria, endotoxins and old RBCs

6 Bile

- Formation and secretion of bile

Gallbladder & Bile

Stores and concentrates bile made by the liver.

GALLBLADDER

- **Parts:** fundus, body, neck
- **Capacity:** 30–50 mL
- **Functions:** stores bile; concentrates bile (active absorption of water & electrolytes)

BILE

- **Composition:** bile salts, bile pigments, cholesterol, lecithin, water, electrolytes
- **Bile salts:** emulsify fat for digestion; form micelles for absorption of fat-soluble vitamins **A, D, E, K**

Calot's Triangle

Also called the cystohepatic triangle. Key landmark in cholecystectomy.

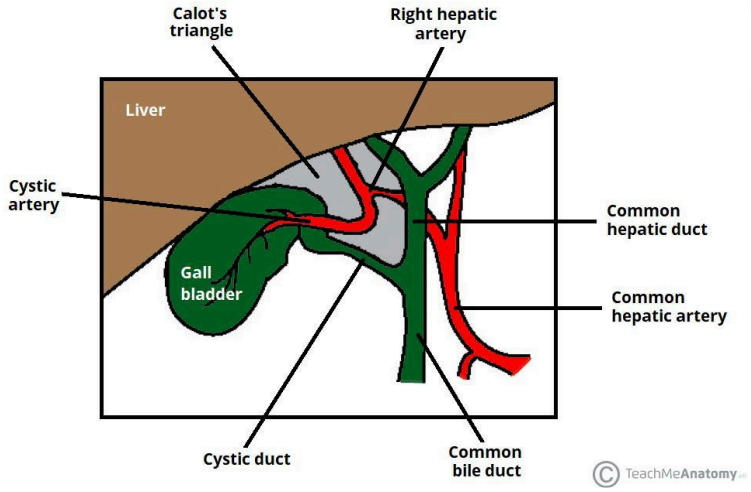


Fig 2. Calot's triangle and its contents.

BOUNDARIES

1. **Cystic duct**: inferolateral
2. **Common hepatic duct**: medial
3. **Inferior edge of liver**: superior

CONTENTS

- Cystic artery
- Cystic lymph node of Lund

KEY Identify Calot's triangle **before clipping the cystic artery** in cholecystectomy, to avoid bleeding and iatrogenic bile duct injury.

Liver Function Tests (LFTs)

A group of biochemical tests that assess **hepatocellular injury**, **synthetic function** and **cholestasis**.

1 · HEPATOCELLULAR INJURY

- **ALT**: more liver specific
- **AST**: also in muscle, heart, RBC
- **AST/ALT ratio >2**: suggests alcoholic liver disease

2 · CHOLESTASIS

- **ALP**: raised in biliary obstruction, PBC
- **GGT**: raised with alcohol, biliary disease

3 · SYNTHETIC FUNCTION

- **Serum albumin**: long half-life, reflects chronic liver disease
- **PT / INR**: short half-life, reflects acute deterioration

4 · EXCRETORY FUNCTION

- **Serum bilirubin**: conjugated vs unconjugated

PART 2 Cirrhosis & Liver Failure

Liver Cirrhosis

End-stage diffuse hepatic injury with **hepatocyte necrosis, fibrosis and regenerative nodules**, disrupting normal liver architecture. Irreversible in most advanced cases.

CAUSES

1. **Viral hepatitis:** HBV, HCV
2. **Alcohol**
3. **NAFLD** (non-alcoholic fatty liver disease)
4. **Autoimmune:** autoimmune hepatitis, primary biliary cholangitis (PBC)
5. **Metabolic:** Wilson disease, haemochromatosis, α-1 antitrypsin deficiency
6. **Vascular:** Budd–Chiari syndrome
7. **Drugs & toxins**

PATHOGENESIS OF FLUID RETENTION IN CIRRHOSIS

Cirrhosis

↓

Portal hypertension → **↑ hydrostatic pressure**

↓ Albumin synthesis → **↓ plasma oncotic pressure**

Splanchnic vasodilatation

Na^+ & water retention (**RAAS activation**)

↓

Ascites & oedema

SCORING SYSTEMS

- **Child–Pugh score:** severity of cirrhosis & surgical risk
- **MELD score:** priority for liver transplant

REMEMBER

- Child–Pugh = **how sick now**
- MELD = **who gets a liver first**

COMPLICATIONS OF CIRRHOSIS

- Portal hypertension
- Variceal haemorrhage
- Ascites
- Spontaneous bacterial peritonitis

Liver cirrhosis

- Hepatic encephalopathy
- Hepatorenal syndrome
- Hepatocellular carcinoma

Portal Hypertension

Persistent elevation of portal venous pressure **>10 mmHg**; clinically significant when **>12 mmHg** (risk of variceal bleeding rises).



PATHOPHYSIOLOGY

1. ↑ **Intrahepatic resistance** (fibrosis & nodule compression)
2. **Splanchnic vasodilatation** → ↑ portal blood flow

OTHER FEATURES

- Caput medusae
- Thrombocytopenia (hypersplenism)

MANAGEMENT

- **Non-selective β-blocker** (propranolol): primary prophylaxis of variceal bleed
- **Endoscopic band ligation** for large varices
- **TIPS** in selected refractory cases

MAIN CLINICAL TRIAD

Splenomegaly + **Ascites** + **Oesophageal varices**

Liver Failure: Acute vs Chronic

	Acute liver failure	Chronic liver failure
Definition	Rapid, severe hepatocyte injury in a patient **without** pre-existing chronic liver disease	Also called chronic liver disease (CLD) or hepatic insufficiency. Progressive deterioration in known cirrhosis; hepatic inflammation & necrosis for **>6 months**

	Acute liver failure	Chronic liver failure
Causes	• **Paracetamol toxicity** • Viral hepatitis A, B, E • Toxins, mushroom poisoning • Ischaemic hepatitis	Any cause of cirrhosis (see Liver cirrhosis)
Key features / complications	• Encephalopathy & cerebral oedema • Metabolic acidosis • Sepsis • Coagulopathy • Multi-organ failure • Electrolytes: hypoglycaemia, hyponatraemia, hypokalaemia	• **Stigmata of CLD:** palmar erythema, spider naevi, gynaecomastia, testicular atrophy, clubbing • Galactorrhoea, amenorrhoea (women) • **Portal hypertension:** ascites, splenomegaly, varices • **Hepatocellular failure:** jaundice, coagulopathy, encephalopathy • Asterixis (flapping tremor) • Complications of ascites: SBP, hepatorenal syndrome

PART 3 Complications of Cirrhosis

Hepatic Encephalopathy (HE)

Neuropsychiatric syndrome due to liver dysfunction / portosystemic shunting; **reversible in early stages**. Asterixis (flapping tremor) is seen.

PATHOGENESIS: AMMONIA IS THE KEY TOXIN

Gut
Protein → ammonia (NH_3)

→

Failing liver
Cannot convert NH_3 → urea; or blood bypasses liver via shunts

→

↑ Blood ammonia
Reaches systemic circulation

→

Brain
Encephalopathy, asterixis

PRECIPITATING FACTORS

1. **GI bleeding** (protein load from blood)
2. **Infection** (SBP, UTI)
3. **Constipation**
4. **Electrolyte imbalance:** hypokalaemia, hyponatraemia
5. **Drugs:** sedatives, antidepressants, diuretics (dehydration)
6. Excess dietary protein
7. Alcohol
8. Portosystemic shunting (TIPS)
9. Large-volume paracentesis
10. Renal failure

Ascites

Accumulation of free fluid within the peritoneal cavity.

TRANSUDATE VS EXUDATE

Feature	Transudative ascites	Exudative ascites
Protein	**Low:** <3 g/dL	**High:** >3 g/dL
Mechanism	↑ Hydrostatic pressure / ↓ plasma oncotic pressure	Inflammation → ↑ peritoneal capillary permeability
Common causes	1. Liver cirrhosis 2. Congestive cardiac failure 3. Nephrotic syndrome 4. Protein-losing enteropathy 5. Hypoalbuminaemia 6. Malabsorption, kwashiorkor	1. Tuberculous peritonitis 2. Malignancy (peritoneal metastasis) 3. Pancreatitis, SLE 4. Bacterial peritonitis 5. Chylous ascites 6. Bile peritonitis 7. Pneumonia, pulmonary infarction

SERUM–ASCITES ALBUMIN GRADIENT (SAAG)

KEY SAAG = serum albumin − ascitic fluid albumin. **≥1.1 g/dL means portal hypertension** is the cause.

	High SAAG (≥1.1 g/dL)	Low SAAG (<1.1 g/dL)
Mechanism	Portal hypertension → splanchnic vasodilatation & fluid shift	Peritoneal inflammation / leakage; no significant portal hypertension
Causes	1. **Hepatic cirrhosis** (most common) 2. Alcoholic hepatitis 3. Cardiac ascites (CCF, constrictive pericarditis) 4. Budd–Chiari syndrome 5. Veno-occlusive disease 6. Massive liver metastasis with portal hypertension	1. Tuberculous peritonitis 2. Peritoneal carcinomatosis 3. Pancreatic ascites 4. Bile ascites 5. Nephrotic syndrome* 6. Serositis (connective tissue disease) 7. Chylous ascites (non-portal cause)

SPONTANEOUS BACTERIAL PERITONITIS (SBP)

Infection of ascitic fluid **without an intra-abdominal source**; common in cirrhosis.

- **Diagnostic:** ascitic fluid PMN **≥250 cells/mm³**
- **Treatment:** empirical third-generation cephalosporin

MANAGEMENT OF CIRRHOTIC ASCITES

- Low-sodium diet
- Diuretics: **spironolactone + furosemide**
- Large-volume paracentesis
- TIPS for refractory ascites

Hepatorenal Syndrome (HRS)

Functional, reversible renal failure occurring in advanced cirrhosis.

Splanchnic vasodilatation → **↓ Effective arterial blood volume** → **Intense renal vasoconstriction** → **Functional renal failure**

DIAGNOSTIC CRITERIA

- Cirrhosis + ascites
- Serum creatinine **>133 µmol/L**
- No response after stopping diuretics + volume expansion
- No shock, nephrotoxins or intrinsic renal disease

TREATMENT

- **Vasoconstrictor (terlipressin) + albumin infusion**
- TIPS in selected patients
- **Definitive:** liver transplantation

KEY HRS is **functional**: the kidneys themselves are normal and can recover after liver transplant.

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PART 4

Jaundice & Hepatitis

Jaundice Types

Where the problem sits on the bilirubin pathway decides the type of jaundice.

RBC breakdown Haem → bilirubin → **Unconjugated** In blood, to liver → **Liver** Uptake, conjugation, secretion → **Bile ducts** Conjugated bilirubin to gut → **Gut** Stercobilinogen (stool), urobilinogen (urine)

Pre-hepatic | Hepatic | Post-hepatic

Parameter	Pre-hepatic	Hepatic (hepatocellular)	Post-hepatic (obstructive / cholestatic)
Primary defect	Excess haemolysis → overproduction of unconjugated bilirubin; liver cannot handle the load	Hepatocyte injury → impaired uptake, conjugation and secretion into bile	Blocked biliary outflow → conjugated bilirubin regurgitates into blood
Bilirubin raised	**↑ Unconjugated**	**↑ Both (mixed)**	**↑ Conjugated**
Urine bilirubin	Absent	Variable	**Present** (dark urine)
Urine urobilinogen	**Increased**	Variable	**Decreased**
Faecal stercobilinogen	**Increased**	**Decreased**	**Decreased**
Stool colour	Dark	Normal or pale	**Pale / clay-coloured**
Liver enzymes	Normal	**Marked ↑ ALT, AST**	**Marked ↑ ALP & GGT**
PT / INR	Normal	Prolonged (synthetic failure)	Prolonged; **improves after vitamin K**
Causes	1. Hereditary spherocytosis 2. G6PD deficiency 3. Sickle cell anaemia 4. Autoimmune haemolytic anaemia 5. Malaria 6. Thalassaemia	1. Acute viral hepatitis 2. Alcoholic hepatitis 3. Drug-induced hepatitis (paracetamol) 4. Cirrhosis 5. Wilson disease, haemochromatosis 6. Ischaemic hepatitis	1. Choledocholithiasis 2. Carcinoma head of pancreas 3. Cholangiocarcinoma 4. Biliary stricture 5. Biliary atresia (paediatrics) 6. PBC 7. Sepsis

Viral Hepatitis A–E

KEY Faeco-oral = **A and E** (acute only, never chronic). Parenteral = **B, C, D** (can become chronic → cirrhosis, HCC).

Virus	Transmission	Acute / chronic	Diseases caused	Key serology
HAV	Faeco-oral	Acute only, never chronic	Acute hepatitis; rarely fulminant failure	**Anti-HAV IgM** = acute **Anti-HAV IgG** = past infection
HBV	Parenteral: blood, needle-stick, IV drug use, sexual, vertical (mother → child)	Acute + chronic	Acute & chronic hepatitis, cirrhosis, HCC, carrier state	**HBsAg** = current infection **Anti-HBs** = vaccinated or recovered **Anti-HBc** = past exposure **HBeAg** = active replication
HCV	Parenteral: blood transfusion, IV drug use, needles	Acute + chronic	Acute & chronic hepatitis, cirrhosis, HCC	**Anti-HCV** = exposure **HCV RNA** = active infection
HDV	Parenteral; **only in HBsAg-positive** patients	Acute + chronic: co-infection usually self-limited; superinfection	Fulminant hepatitis, rapidly progressive cirrhosis, HCC	**Anti-HDV IgM** = acute **HDV RNA** = replication Always with HBsAg +
HEV	Faeco-oral	Acute only, no chronic	Acute hepatitis; **fulminant failure in pregnancy**	**Anti-HEV IgM** = acute **Anti-HEV IgG** = past exposure

Drug-Induced Hepatitis

Liver injury caused by medications, herbal or complementary drugs.

CAUSES

- **Paracetamol (acetaminophen): most common**
- Isoniazid
- Rifampicin
- Pyrazinamide
- Amoxicillin–clavulanate
- Statins
- Phenytoin

Gallstones (Cholelithiasis)

Pathogenesis: cholesterol supersaturation of bile and gallbladder stasis.

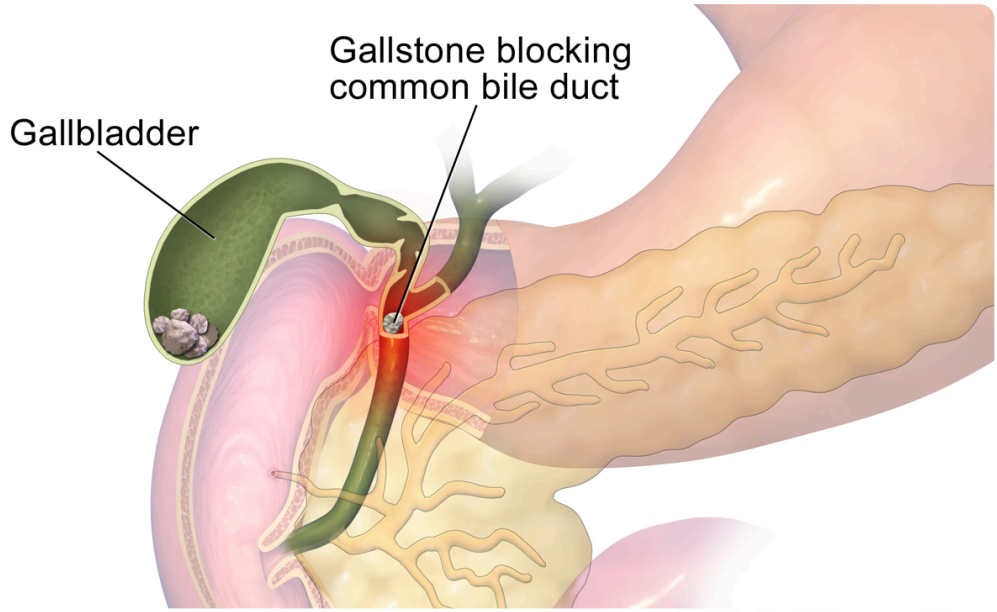


Fig 3. Gallstones in the gallbladder and a stone blocking the common bile duct.

Image: BruceBlaus (Blausen Medical), CC BY-SA 4.0, via Wikimedia Commons; cropped.

RISK FACTORS · THE 4 FS

Female | Forty | Fat | Fertile

- Obesity, rapid weight loss
- Multiparity, oral contraceptive pill

TYPES

1. **Cholesterol stones** (most common)
2. **Pigment stones:** black (haemolysis), brown (biliary infection)

CLINICAL PRESENTATION

- Majority are **asymptomatic**
- **Biliary colic:** severe episodic right upper quadrant pain after a fatty meal

COMPLICATIONS

- Acute cholecystitis
- Choledocholithiasis
- Acute pancreatitis
- Gallbladder carcinoma

TREATMENT

- **Asymptomatic:** no surgery
- **Symptomatic:** laparoscopic cholecystectomy

Ascending Cholangitis

Bacterial infection of the biliary tree, **secondary to biliary obstruction.**

OBSTRUCTION

- **Choledocholithiasis** (most common)
- Stricture, biliary stent, tumour

ORGANISMS

E. coli | *Klebsiella* | *Enterococcus*

CHARCOT'S TRIAD (CLASSIC)

Jaundice + Right upper quadrant pain + Fever with chills

REYNOLDS' PENTAD (SEVERE / SUPPURATIVE CHOLANGITIS)

Charcot's triad + Shock + Altered mental status

PART 6 Other Liver Diseases

Budd–Chiari Syndrome

Hepatic venous outflow obstruction.

CAUSES

- Myeloproliferative disorders (polycythaemia vera)
- Thrombophilia (factor V Leiden)
- Pregnancy, oral contraceptives, malignancy

CLINICAL TRIAD

Ascites | Painful hepatomegaly | Jaundice

Hepatocellular Carcinoma

RISK FACTORS

1. **Chronic hepatitis B** (#1 cause in Bangladesh)
2. Chronic hepatitis C
3. Cirrhosis
4. Aflatoxin (contaminated food)
5. Alcohol, NAFLD

SCREENING

USG abdomen + **alpha-fetoprotein (AFP) every 6 months** in cirrhotic patients

TREATMENT

Resection, radiofrequency ablation, liver transplant

Wilson's Disease

Autosomal recessive **ATP7B** gene mutation

↓

↓ Copper excretion by liver

↓

Copper builds up in **liver, brain, cornea**

CLINICAL FEATURES

- **Liver:** hepatitis, cirrhosis, acute liver failure
- **Neurological:** tremor, dysarthria, dystonia
- Psychiatric features
- **Kayser–Fleischer ring** (corneal copper)

INVESTIGATIONS

- ↓ Serum ceruloplasmin
- ↑ 24-hour urinary copper
- ↑ Hepatic copper on liver biopsy

TREATMENT

Chelating agent: **penicillamine**

Haemochromatosis

Autosomal recessive

↓

↑ Intestinal iron absorption

↓

Iron deposits in **liver, pancreas, heart, skin, joints**

CLASSIC PRESENTATION: "BRONZE DIABETES"

Skin pigmentation | Diabetes mellitus | Cirrhosis

OTHER FEATURES

Arthropathy, cardiomyopathy, hypogonadism

INVESTIGATIONS

- ↑ Serum ferritin
- ↑ Transferrin saturation (>45%)
- Liver biopsy: iron overload

TREATMENT

- **Regular phlebotomy**: first line
- Iron chelators